

Answer **all** the questions.

1. A circular patch of oil is forming on the surface of the sea due to an oil leak. At 08 00 hours, the radius of the patch is 40 m and the area of the patch is increasing at a rate of 250 m^2 per hour. Find the rate of increase of the radius at 08 00 hours. [3]

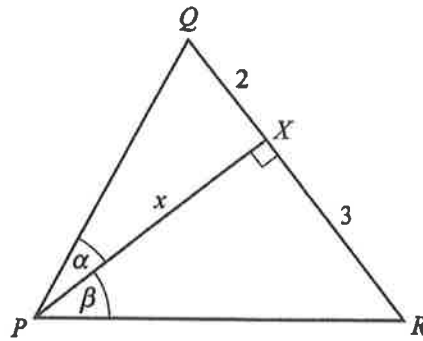
2. (a) Find $\frac{d}{dx}\sqrt{x^2 + x + 1}$.

[2]

(b) Hence find $\int \frac{16x + 8}{\sqrt{x^2 + x + 1}} dx$.

[2]

3.



The diagram shows a triangle PQR . Point X lies on QR such that $QX = 2$ units and $XR = 3$ units. Angle $PXR = 90^\circ$ and PX is x units. Angles QPX and RPX are denoted by α and β as shown. Given that $\tan(\alpha + \beta) = 2$, find the value of x .

[5]

4. The line $2y + x = 1$ cuts the curve $x^2 + y^2 - 2x = 4$ at the points A and B . Find the coordinates of the midpoint of AB . [6]

5. The term independent of x in the expansion of $\left(2x - \frac{b}{x}\right)^6$, where b is a constant, is -540 .

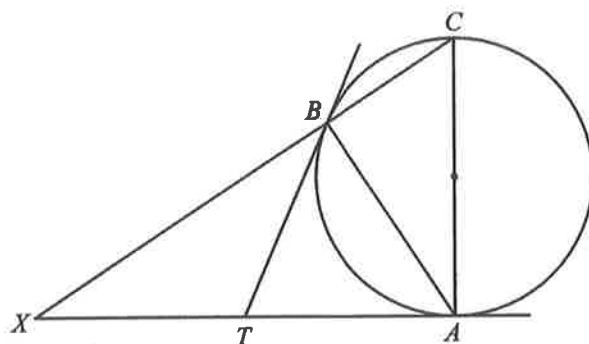
(a) Find the value of b .

[4]

(b) Using your value of b , show that there is no term in x^2 in the expansion of $(1 + x^2)\left(2x - \frac{b}{x}\right)^6$.

[3]

6.



In the diagram, A , B and C lie on a circle and AC is a diameter. The tangents to the circle at A and B meet at T . The line CB extended meets the line AT extended at X . Show, giving all reasons, that $\angle ATB = 2 \times \angle AXB$. [6]

7. A solid cuboid has a square base of side $(2 + \sqrt{2})$ cm. The volume of the cuboid is $(14 + 8\sqrt{2})$ cm³. **Without using a calculator**, express the total surface area of the cuboid in the form $(a + b\sqrt{2})$ cm², where a and b are constants. [7]

8. A rectangular cardboard box has a square base of side x cm and a height of y cm. The total surface area of the six faces of the box is 600 cm^2 .

(a) Show that the volume of the box, $V \text{ cm}^3$, is given by $V = 150x - \frac{x^3}{2}$. [3]

Given that x can vary,

(b) find the stationary value of V and determine its nature.

[4]

9. (a) (i) Express $2x^2 - 6x + 7$ in the form $a(x + b)^2 + c$ where a , b and c are constants. [2]
- (ii) Hence, or otherwise, find the range of values of the constant k for which the equation $2x^2 - 6x + 7 = k$ does not have two distinct roots. [2]
- (b) The equation of a curve is $y = p(x - q)^2 + r$ where p , q and r are constants. The curve intersects the x -axis at $(2, 0)$ and $(4, 0)$ and the maximum value of y is 2. Find the values of p , q and r . [4]

10. (a) State the amplitude and period of $4 \cos 2x$.

[2]

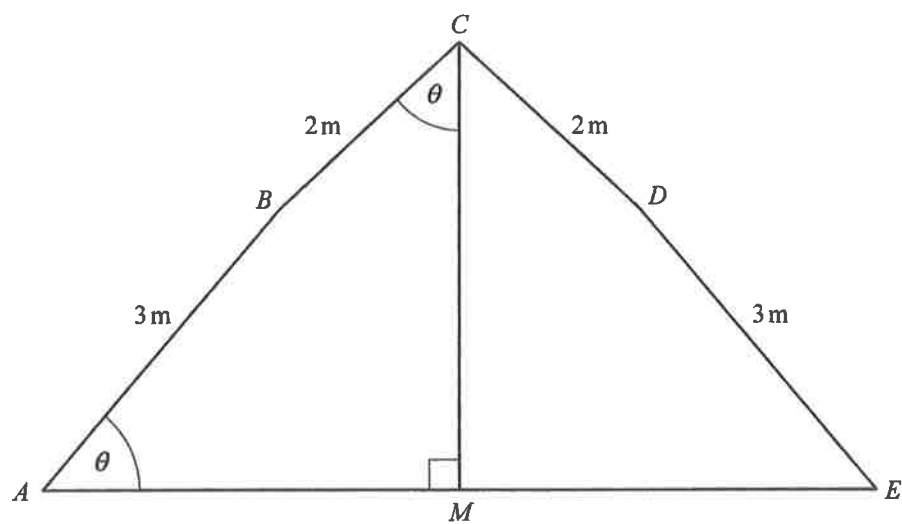
(b) Sketch the graph of $y = 4 \cos 2x$ for $0 \leq x \leq 2\pi$.

[3]

(c) By drawing a suitable straight line on your sketch, determine the number of solutions of the equation $\pi \cos 2x + x = 0$.

[3]

11.



(a) Show that the length CM can be expressed as $2 \cos \theta + 3 \sin \theta$.

[2]

- (b) By expressing $2 \cos \theta + 3 \sin \theta$ in the form $R \cos (\theta - \alpha)$ where $R > 0$ and α is acute, find the maximum possible length of CM . [5]

- (c) Find the value of θ for which $CM = 3.5$ m. [2]

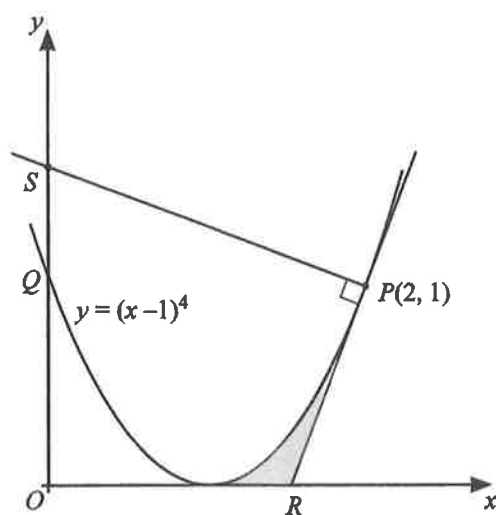
12. (a) Express $\frac{3x-5}{x^2(x-1)}$ in partial fractions.

[5]

(b) Use your answer to **part (a)** to show that $\int_2^3 \frac{3x-5}{x^2(x-1)} dx = \frac{5}{6} - \ln \frac{16}{9}$.

[5]

13.



The diagram shows part of the curve $y = (x - 1)^4$ cutting the y -axis at Q . The point $P(2, 1)$ lies on the curve. The tangent to the curve at P cuts the x -axis at R and the normal to the curve at P cuts the y -axis at S .

(a) Determine, with full working, whether Q is nearer to S or to O .

[4]

(b) Find the area of the shaded region.

[6]

